



UGC & Govt. Approved
Sonargaon University (SU)
সোনারগাঁও ইউনিভার্সিটি (এসইউ)

Course Title: Financial Management
Course Code: FIN312 / FIN533

LECTURE SHEET

Lectured By:
Md. Shahbub Alam
Lecturer, Dept. of Business Administration
Sonargaon University (SU)

Chapter: Stock Valuation

Questions at a glance:

1. Describe the main features of the common stock.
2. Describe the main advantages of common stock as investment.
3. What is meant by the intrinsic (investment) value of a stock?

1. Describe the main features of the common stock.

The main features of the common stock are described as follows:

- a) Typically each common stock owned entitles an investor to one vote in corporate shareholders' meeting.
- b) Investor receives benefits in the form of dividends, capital gains or both. But:
 - Dividends are paid to shareholders only after other liabilities such as interest payments have been settled.
 - Typically the firm does not pay all its earnings in cash dividends;
 - Special form of dividend is stock dividend, in which the corporation pays in stocks rather than cash.
- c) Common stock has no stated maturity. Common stock does not have a date on which the corporation must buy it back. But some corporations pay cash to their shareholders by purchasing their own shares. These are known as share buybacks.
- d) Common stocks on the whole historically have provided a higher return, but they also have higher risk. An investor earns capital gains (the difference between the purchase price and selling price) when he/she sell at a higher price than the purchase price.

2. Describe the main advantages of common stock as investment.

The main advantages of the common stock are described as follows:

- a) The investment income is usually higher.
- b) The investor can receive operating income in cash dividends.
- c) Common stock has a very high liquidity and can easily be moved from one investor to the other.
- d) The costs of transaction with common stocks involved are relatively low.
- e) The nominal price of common stock is lower in comparison with the other securities.

3. What is meant by the intrinsic (investment) value of a stock?

The intrinsic value is also known as real or fire value. The intrinsic value of a stock represents the present value of its future of its future earnings discounted at an appropriate or required rate of return.

Intrinsic value of a stock can be calculated as follows:

$$V = \text{Present value of dividends} + \text{Present value of stock price}$$

The intrinsic value of stock depends upon the earnings and dividends as also on the capitalization rate or discount rate which represents the minimum rate expected by investors to buy or hold a stock.

Formula

- **Preference Share Valuation**

$$P_0 = \frac{D_p}{r}$$

Where,

P_0 = value of preference share at time zero

D_p = dividend payment

r = required rate of return

- **Equity Valuation**

➤ Using zero dividend growth model:

$$P_0 = \frac{D_1}{r}$$

Where,

P_0 = value of equity share at time zero

D_1 = dividend payment expected one year from now = D_0

r = required rate of return

➤ Using constant dividend growth model:

$$P_0 = \frac{D_1}{r - g}$$

Where,

P_0 = value of equity share at time zero

g = growth rate

D_1 = dividend payment expected one year from now = $D_0(1 + g)$

r = required rate of return

➤ Using variable dividend growth model:

Suppose, growth rate g_1 is for next 3 years and growth rate g_2 is for 4th year to α .

Step 1: The values of cash dividend in the each of the next 3 year are as follows:

$$D_1 = D_0 (1 + g_1)$$

$$D_2 = D_1 (1 + g_1)$$

$$D_3 = D_2 (1 + g_1)$$

$$\text{PV of dividends} = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3}$$

Step 2: The value of stock at the end of the years 3 is calculated as follows:

$$P_3 = \frac{D_4}{r - g_2} \quad \text{Here, } D_4 = D_3 (1 + g_2)$$

$$\text{PV of } P_3 = \frac{P_3}{(1+r)^3}$$

Step 3: The value of common stock today:

$$P_0 = \text{PV of dividends} + \text{PV of } P_3$$

Examples

1. Louise Marshall owns 1,000 £1.00 par value 5 per cent preference shares in ABC plc and now wishes to determine the value of her investment. The required rate of return on similar shares is currently 6 per cent. Calculate the value of preference share.

Answer:

Given that,

$$D_p = (1,000 \times £1.00) \times 5\% = £50$$

$$r = 6\% = 0.06$$

$$P_0 = \frac{D_p}{r} \\ = \frac{£50}{0.06} = £833.33$$

2. Sunny Sky Ventures is expected to pay a dividend of £0.20 per share indefinitely; assuming the investor's required rate of return is 12 per cent, calculate the value of the share.

Answer:

Given that,

$$D_1 = £0.20$$

$$r = 12\% = 0.12$$

$$P_0 = \frac{D_1}{r} \\ = \frac{£0.20}{0.12} = £1.67$$

3. Sunny Sky Ventures has just paid a current dividend for the year of £0.20 per share and the company expects future dividends to grow at a constant 5 per cent per year. Assuming the investor's required rate of return is 12 per cent; calculate the value of the share.

Answer:

Given that,

$$g = 5\% = 0.05$$

$$D_0 = £0.20$$

$$\begin{aligned} \therefore D_1 &= D_0(1 + g) \\ &= £0.20(1 + 0.05) \\ &= £0.21 \end{aligned}$$

$$r = 12\% = 0.12$$

$$P_0 = \frac{D_1}{r - g} \\ = \frac{£0.21}{0.12 - 0.05} = \frac{£0.21}{0.07} = £3.00$$

4. Set out below is the dividend per share (DPS) paid by Meteoric Enterprises over the period 1994 to 1999.

<u>Year</u>	<u>Dividend per Share (DPS)</u>
1999	£0.50
1998	£0.42
1997	£0.35
1996	£0.29
1995	£0.24
1994	£0.20

Use this information to calculate the historic growth rate in dividend per share.

Answer:

We know that,

$$D_1 = D_0(1 + g)$$

Substituting,

$$D_{1999} = D_{1994} (1 + g)^5$$

$$\text{or, } £0.50 = £0.20 (1 + g)^5$$

$$\text{or, } £0.50/£0.20 = (1 + g)^5$$

$$\text{or, } (1 + g)^5 = 2.5$$

$$\text{or, } 1 + g = \sqrt[5]{2.5}$$

$$\text{or, } 1 + g = 1.20$$

$$\text{or, } g = 0.20$$

$$\therefore g = 20\%$$

5. Prestige Packaging has paid the annual dividends per share as set out below. The company's directors expect that the past dividend growth rate will also be maintained at a constant rate in the future. The company has just paid its dividend for 1999 of £0.35 per share. If investors currently require a share of equal risk to earn a return of 15 per cent, calculate the value of the share.

<u>Year</u>	<u>Dividend per Share (DPS)</u>
1999	£0.35
1998	£0.33
1997	£0.31
1996	£0.29
1995	£0.27
1994	£0.25

Briefly state any assumptions made in using this model to value shares.

Answer:

We know that,

$$D_1 = D_0(1 + g)$$

Substituting,

$$D_{1999} = D_{1994} (1 + g)^5$$

$$\text{or, } £0.35 = £0.25 (1 + g)^5$$

$$\text{or, } £0.35/£0.25 = (1 + g)^5$$

$$\text{or, } (1 + g)^5 = 1.4$$

$$\text{or, } 1 + g = \sqrt[5]{1.4}$$

$$\text{or, } 1 + g = 1.07$$

$$\text{or, } g = 1.07 - 1$$

$$\therefore g = 0.07$$

Here,

$$g = 0.07$$

$$D_0 = £0.35$$

$$\therefore D_1 = D_0(1 + g)$$

$$= £0.35 (1 + 0.07)$$

$$= £0.375$$

$$r = 15\% = 0.15$$

$$\begin{aligned}\text{Now, } P_0 &= \frac{D_1}{r - g} \\ &= \frac{£0.375}{0.15 - 0.07} \\ &= £4.69\end{aligned}$$

The model assumes that the future growth rate will replicate the past growth rate, and that the required rate of return has been accurately determined.

6. Perry Motors' common stock just paid its annual dividend of \$1.80 per share. The required return on the common stock is 12%. Estimate the value of the common stock under each of the following assumptions about the dividend:
- Dividends are expected to grow at an annual rate of 0% to infinity.
 - Dividends are expected to grow at a constant annual rate of 5% to infinity.
 - Dividends are expected to grow at an annual rate of 5% for each of the next 3 years, followed by a constant annual growth rate of 4% in years 4 to infinity.

Answer:

Given that,

$$D_0 = \$1.80$$

$$r = 12\% = 0.12$$

a. Zero Growth:

$$\begin{aligned} P_0 &= \frac{D_1}{r} \\ &= \frac{\$1.80}{0.12} \\ &= \$15 \end{aligned}$$

Here,

$$D_1 = D_0 = \$1.80$$

b. Constant Growth:

$$\begin{aligned} P_0 &= \frac{D_1}{r - g} \\ &= \frac{\$1.89}{0.12 - 0.05} \\ &= \frac{\$1.89}{0.07} \\ &= \$27 \end{aligned}$$

Here,

$$g = 5\% = 0.05$$

$$\begin{aligned} D_1 &= D_0(1 + g) \\ &= \$1.80(1 + 0.05) = \$1.89 \end{aligned}$$

c. Variable Growth:

Given that,

$$g_1 = 5\% \text{ for year 1 to 3}$$

$$g_2 = 4\% \text{ for year 4 to } \alpha$$

Step 1: The values of cash dividend in the each of the next 3 year are calculated as follows:

$$D_1 = D_0(1 + g_1) = \$1.80(1 + 0.05) = \$1.89$$

$$D_2 = D_1(1 + g_1) = \$1.89(1 + 0.05) = \$1.98$$

$$D_3 = D_2(1 + g_1) = \$1.98(1 + 0.05) = \$2.08$$

$$\begin{aligned} \text{PV of dividends} &= \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} \\ &= \frac{\$1.89}{(1+0.12)^1} + \frac{\$1.98}{(1+0.12)^2} + \frac{\$2.08}{(1+0.12)^3} \\ &= \$1.69 + \$1.58 + \$1.48 \\ &= \$4.75 \end{aligned}$$

Step 2: The value of stock at the end of the years 3 is calculated as follows:

$$\begin{aligned} P_3 &= \frac{D_4}{r - g_2} \\ &= \frac{\$2.16}{0.12 - 0.04} \\ &= \$27 \end{aligned}$$

Here,

$$\begin{aligned} D_4 &= D_3(1 + g_2) \\ &= \$2.08(1 + 0.04) \\ &= \$2.16 \end{aligned}$$

$$\begin{aligned} \text{PV of } P_3 &= \frac{P_3}{(1+r)^3} \\ &= \frac{\$27}{(1+0.12)^3} \\ &= \$19.22 \end{aligned}$$

Step 3: The value of common stock today:

$$\begin{aligned}P_0 &= \text{PV of dividends} + \text{PV of } P_3 \\&= \$4.75 + \$19.22 \\&= \$ 23.97\end{aligned}$$

Self-Test Problems

Managerial Finance - L.J. Gitman, 13th Edition (Page No: 298 & 299)

Problems No: P7–10, P7–11, P7–12, P7–13 & P7–14.